Nathan Shankar

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ABOUT ME

PhD researcher specializing in robot navigation in the dark using vision. Proven ability in designing and developing intelligent algorithms for optimizing robot performance, with a focus on robotic navigation, computer vision, and machine learning. Demonstrated expertise in leading innovative projects from concept through execution, enhancing system functionalities and user experience in robotic systems. Committed to pushing the boundaries of robot perception and navigation to solve real world challenges.

DIGITAL SKILLS

ROS & ROS2 / Gazebo (Robotics simulator) / Machine Learning / Fundamental Electronics / C++ / Python / Git / Linux / MATLAB / Docker / Dynamic simulation / SLAM (Simultaneos Localization and Mapping) / Path Planning Algorithms / State estimation / Machine Learing / Autonomous Navigation / IoT / Sensor Fusion / Control systems / Solidworks

EDUCATION AND TRAINING

PhD Mechanical Engineering

University of Manchester [09/2024 – Current]

City: Manchester | **Country:** United Kingdom | **Level in EQF:** EQF level 8 | **Thesis:** Autonomous Navigation for Robots in dark unknown environments.

MSc Robotics

University of Manchester [09/2023 – 07/2024]

City: Manchester | **Country:** United Kingdom | **Final grade:** Distinction | **Level in EQF:** EQF level 7 | **Thesis:** IoT based path planner for UAVs

- Software for Robotics
- Robotic Systems
- Robotic Manipulators
- Cognitive Robotics and Computer Vision
- Autonomous Mobile Robots
- Foundations of Machine Learning
- Robotic Systems Design Project

BTech Electrical and Electronics Engineering

Vellore Institute of Technology [06/2019 – 08/2023]

City: Vellore | **Country:** India | **Final grade:** 9.49 CGPA | **Level in EQF:** EQF level 6

- Robotics
- Control Systems
- Internet of Things
- Electric Vehicles
- Network Theory
- Power Systems
- Power Electronics and Drives

WORK EXPERIENCE

Graduate Teaching Assistant - Autonomous Mobile Robots

University of Manchester [01/2025 – Current]

City: Manchester | **Country:** United Kingdom

- Providing students with guidance and technical support on course materials and study skills, and referring them to appropriate support for pastoral care.
- Teaching seminars and tutorials.
- Marking and providing feedback on students' coursework.
- Undertaking laboratory demonstrations and support activities during practical sessions.

Intern

Bodhi Labs Software Services Pvt Ltd [12/2022 – 04/2023]

City: Bengaluru | **Country:** India

- Led assembly and configuration of semi-autonomous table-tennis robot, enhancing mechanical and electronic systems.
- Collaborated in brainstorming sessions to solve client issues, driving innovative solutions.
- Executed rigorous testing to ensure robot safety and improve user experience.
- Programmed Arduino to boost robot functionality, integrating application endpoints with cross-functional teams.
- Developed promotional content highlighting the physics of table-tennis, increasing product visibility.
- Refined software models by analyzing user feedback, boosting functionality.

Projects Manager

Students for The Exploration and Development of Space (SEDS India) [09/2021 – 09/2022]

Country: India

- Conducted advanced lectures on State Estimation, Path Planning, and SLAM, enhancing student knowledge.
- Guided student chapters in project team formation and lifecycle support, ensuring project success.
- Managed rover team for University Rover Challenge, leading to successful event participation.
- Designed and optimized rover for Mars soil excavation, reducing costs and improving manufacturability.
- Presented rover design reviews using Solidworks, performing dynamic simulations and developing ROS skills.

Projects Manager

IEEE Circuits and System - University Chapter [09/2020 – 09/2021]

City: Vellore | **Country:** India

- Led project teams in signal processing, delivering innovative algorithms using MATLAB and Python.
- Supervised PCB design and circuit simulations, enhancing project outcomes with Arduino and sensors.
- Encouraged team creativity, resulting in novel problem-solving approaches and measurable project success.

PROJECTS

[11/2023 – 03/2024]

Leo Rover

Worked in a team of 4 on a fully autonomous AGV integrated with Raspberry Pi and a custom microcontroller capable of self-navigating an unmapped terrain and retrieving an object of interest. Implemented autonomous navigational capabilities for the robot. Worked on creating software packages, system integration, SLAM, sensor integration, electrical subsystem, and control dashboard for the rover.

Link: https://github.com/nathanshankar/leo-rover-autonomous-exploration

[03/2024 – 04/2024]

Drone Control

My controller utilizes PID tuning for the y-axis, x-axis, and attitude, emphasising speed and stability while accommodating different error conditions using the Dynamic PID. The integration of a DOBC enhances robustness, effectively countering external disturbances. The controller's blend of traditional and advanced control methods showcases a comprehensive approach to drone control.

Link: https://github.com/nathanshankar/drone-control